Risk Management Sketchbook

11 classes · Risk Management · Fall 2026 finals

Each class: the idea in one line, the sketch step by step, plain explanation, formulas, a worked example, common traps and where to read more.

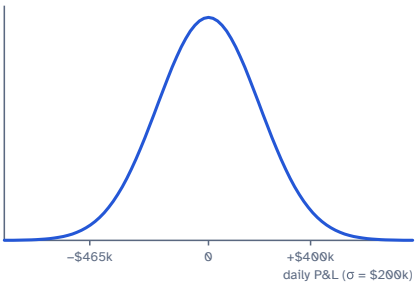
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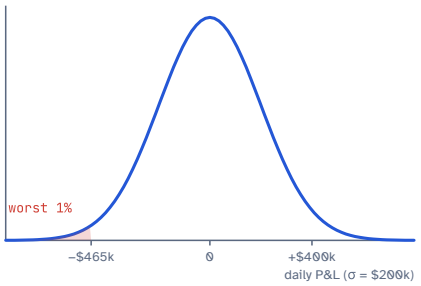
Volatility, diversification, VaR and ES

VaR answers "how bad is a bad day?" Expected Shortfall answers "and if it gets worse than that, how bad on average?"

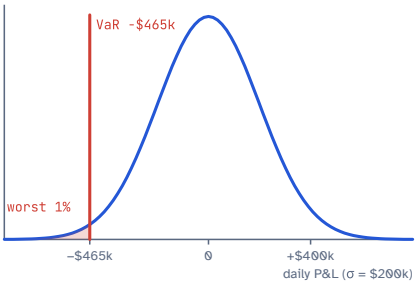
The sketch, step by step



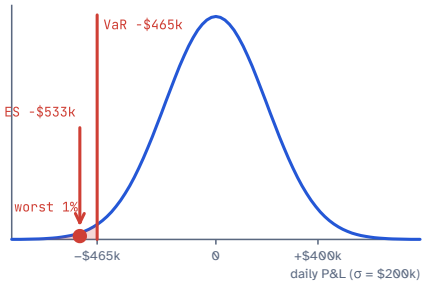
Step 1. This bell curve is your daily profit and loss. Most days are near zero.



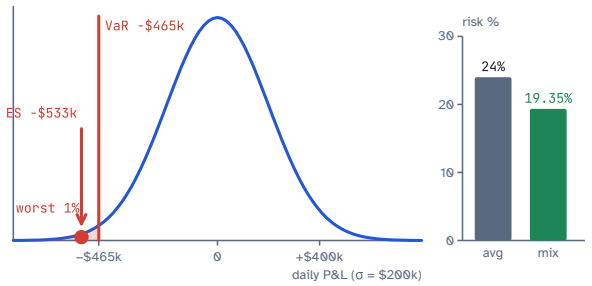
Step 2. The red sliver on the left is the worst 1% of days.



Step 3. VaR is where that sliver starts: a loss of $465k.



Step 4. ES is the average of everything inside the sliver: $533k. Further out than VaR.



Step 5. Diversification: mixing two assets gives 19.35% risk, less than the 24% average.

What it means

- **Volatility (σ):** the standard deviation of returns. Bigger σ = wider swings.
- **Diversification:** two assets that don't move together have a portfolio σ *below* the average of their σs. Correlation ρ controls how much.
- **VaR at 99%:** the loss you will not exceed on 99 out of 100 days. For a normal distribution it is 2.326 × σ.
- **Expected Shortfall (ES):** the average loss on the worst 1% of days. Always bigger than VaR, and it sees how fat the tail is. Regulators now prefer ES.

The formulas

```
Portfolio variance: σp² = w1²σ1² + w2²σ2² + 2·w1·w2·ρ·σ1·σ2
Normal VaR:  VaR = z × σ × Value      (z = 1.645 at 95%, 2.326 at 99%)
Normal ES:   ES  = σ × Value × φ(z) / (1 - confidence)
```

w = weights, ρ = correlation, φ = the normal bell-curve height.

Worked example

Question. A $10m portfolio has a daily σ of 2%. Find the 99% 1-day VaR and ES. Then: 60/40 in two assets with σ = 20% and 30%, ρ = 0.3. Portfolio σ?

1. Daily $σ = 0.02 × 10m = `$200,000`
2. VaR = 2.326 × 200,000 = `$465,000`
3. ES = 200,000 × φ(2.326)/0.01 = 200,000 × 2.665 = `$533,000`
4. $σp^2 = 0.36(0.04) + 0.16(0.09) + 2(0.6)(0.4)(0.3)(0.2)(0.3) =$ `0.03744`, σp = `19.35%`, below the 24% weighted average.

Answer: VaR $465k, ES $533k; portfolio σ 19.35%

Calculator example

Inputs: Portfolio value ($) = 10000000, Daily volatility (%) = 2, Confidence (95 or 99) = 99, Days = 1

```
$σ over 1 day(s) = 200,000.00
VaR = 2.3263 × 200,000.00 = $465,260.00
ES = 200,000.00 × φ(2.3263)/0.01 = $533,102.21
```

Common traps

- Calling VaR the worst possible loss. It is a threshold, not a maximum.
- Scaling 1-day VaR to 10 days by 10 instead of √10.
- Forgetting the 2·w1·w2 in the variance formula.

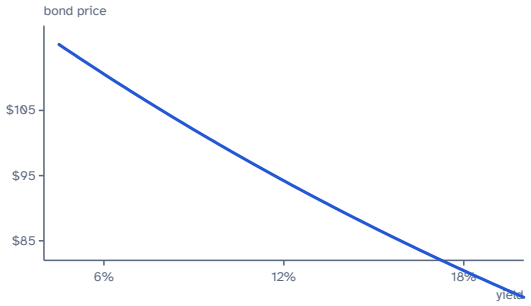
Where to read more

- `Risk/01 Course Information/FRE6123_INFO_01_Syllabus_Fall_2026.docx (RMN1)`
- `Risk lecture decks RMN1-2 are not in your folder yet`

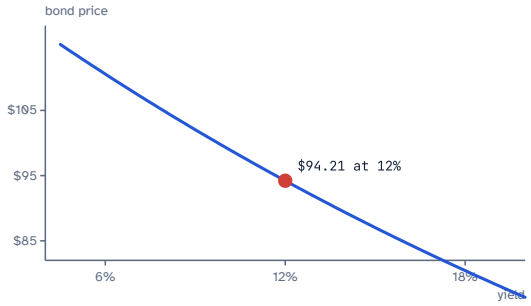
Forward rates, duration and convexity

Duration tells you how much a bond price drops when rates rise. Convexity is the correction, because the real price curve bends.

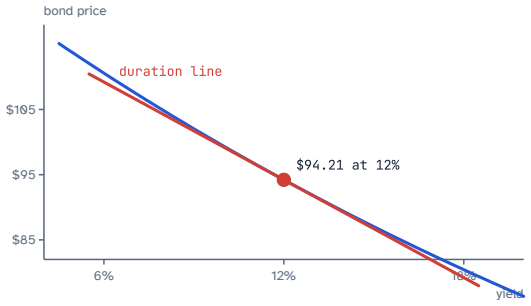
The sketch, step by step



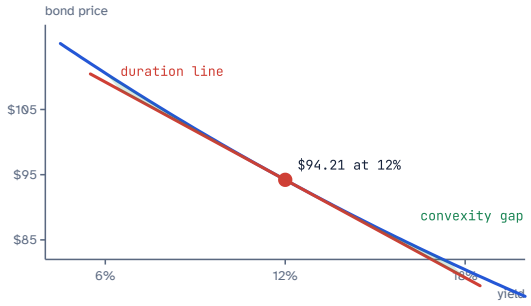
Step 1. The price of the bond at each yield. It slopes down and bends (is "convex").



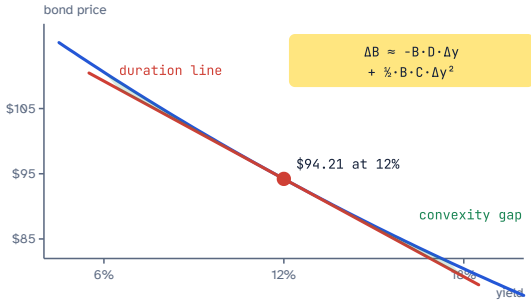
Step 2. At today's yield of 12%, the price is $94.21.



Step 3. Duration draws the tangent line at that point. Slope = −B × D.



Step 4. The tangent is below the curve on both sides. That gap is convexity.



Step 5. For a small move the line is close. For a big move, add the convexity term or reprice fully.

What it means

Forward rate: the rate for a future period that today's zero rates lock in. If 1 year pays 3% and 2 years pays 4%, the second year alone must pay 5% (continuous rates).

Duration is the weighted average time you wait for your cash, where the weights are each payment's share of the price. It is also the sensitivity: a 1% rise in yield cuts the price by about D%.

Convexity: the price-yield curve is bowed. A straight-line (duration) estimate always *underestimates* the price after a big move either way. Full repricing is the true answer; duration

is the quick estimate.

The formulas

```
Forward rate:   F(T1,T2) = (R2·T2 - R1·T1) / (T2 - T1)
Duration:       D = Σ t × [CF(t)·e^(-y·t)] / B
Price change:   ΔB ≈ -B × D × Δy  (+ ½ × B × C × Δy²)
```

R = zero rates, y = bond yield (continuous), B = bond price, C = convexity.

Worked example

Question. 3-year bond, 10% coupon paid twice a year, yield 12% continuous (Hull's example). Price, duration, and the effect of a 0.1% rise in yield?

1. Cash flows: $5 at 0.5, 1, 1.5, 2, 2.5 years and $105 at 3 years.
2. Discount each at e^(−0.12t) and add: `B = 94.213`
3. Weight each time by its share of the price and add: `D = 2.653 years`
4. ΔB ≈ −94.213 × 2.653 × 0.001 = `-0.250`, so the price falls to about 93.963.

Answer: B = 94.21, D = 2.653, price drops about $0.25

Common traps

- Using Δy = 0.1 instead of 0.001.
- Forgetting the minus sign: rates up means price down.
- Using duration for a large rate shock without the convexity term.

Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx`
- `Risk syllabus RMN2 (duration, convexity, term structure)`

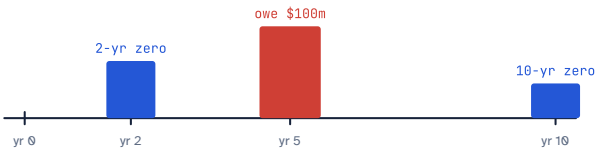
Immunization and asset-liability management

A pension fund or bank owes money in the future. Immunization picks assets whose value moves the same way as those debts when rates change, so a rate move does not hurt.

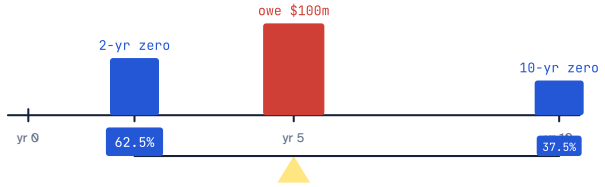
The sketch, step by step



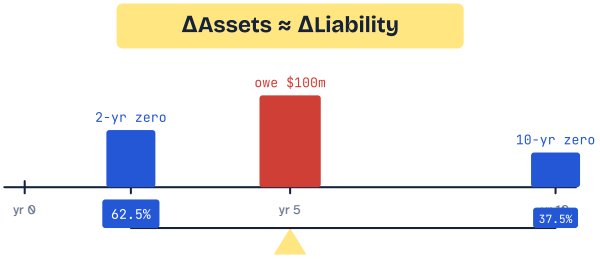
Step 1. A timeline. You owe $100m in present-value terms at year 5.



Step 2. You can only buy bonds that pay at year 2 and year 10.



Step 3. Think of a seesaw balanced at year 5: 62.5% at year 2 and 37.5% at year 10 balance it.



Step 4. Now if rates move a little, assets and the liability move by about the same amount. You are immunized.

What it means

- **ALM (asset-liability management):** manage assets and liabilities together, not separately. The risk is the *gap* between them.
- **Duration matching:** make the PV of assets equal the PV of liabilities, and their durations equal. Then a small parallel rate shift changes both by about the same amount.
- **Convexity:** also have asset convexity ≥ liability convexity, so bigger moves help rather than hurt.
- **Rebalance:** durations drift as time passes and rates move, so immunization must be redone.
- **Key-rate duration:** the curve does not always shift in parallel. Key-rate durations measure exposure to each part of the curve (2y, 5y, 10y…) separately.
- **Hedge accounting** (RMN8): rules that let a company show a hedge and the item it hedges together in its accounts, so the P&L does not jump around.

The formulas

```
Match PV:        PV(assets) = PV(liabilities)
Match duration:  w1·D1 + w2·D2 = D_L,   w1 + w2 = 1
Surplus change:  ΔS ≈ -(A·D_A - L·D_L) × Δy
```

w = share of money in each bond, D = duration, A and L = PV of assets and liabilities.

Worked example

Question. You owe $100m (PV) in 5 years. You can buy 2-year and 10-year zero-coupon bonds. How much of each?

1. Zero-coupon bond duration = its maturity: D1 = 2, D2 = 10.
2. Solve w1 × 2 + w2 × 10 = 5 with w1 + w2 = 1.
3. w2 × 8 = 3 → w2 = `0.375`, w1 = `0.625`
4. Buy $62.5m of 2-year zeros and $37.5m of 10-year zeros (in PV). This "barbell" also has more convexity than the liability.

Answer: 62.5% in 2-year, 37.5% in 10-year

Calculator example

Inputs: Liability duration = 5, Bond 1 duration = 2, Bond 2 duration = 10, PV of liability ($m) = 100

```
w2 = (5 - 2) / (10 - 2) = 0.3750, w1 = 0.6250
Buy $62.50m of bond 1 and $37.50m of bond 2 (PV)
```

Common traps

- Matching maturities instead of durations.
- Forgetting to also match PV.
- Assuming the hedge lasts forever; durations drift and must be rebalanced.

Where to read more

- `Risk syllabus RMN2 and RMN8 (duration, immunization, ALM, hedge accounting)`
- `Derivatives/03 Lectures/FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx`

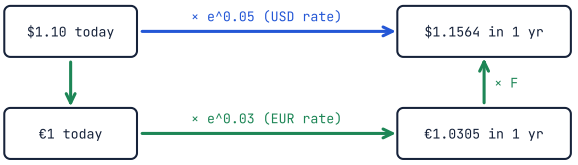
Currency and commodity risk

Exchange-rate forwards follow interest rate parity: the currency with the higher interest rate trades at a discount forward. Commodity forwards add storage costs and a convenience yield.

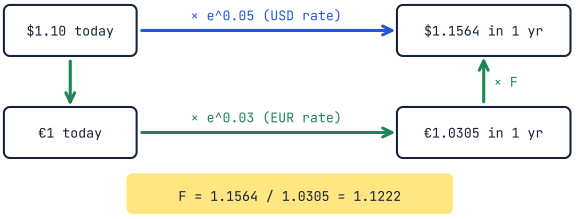
The sketch, step by step



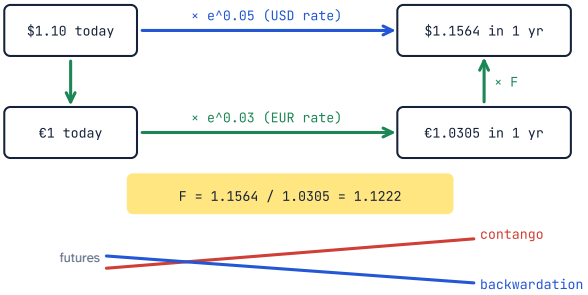
Step 1. Path 1: keep $1.10 in dollars at 5% for a year. You end with 1.10 × e^0.05.



Step 2. Path 2: change it to €1, earn 3% in euros, and convert back at the forward rate F.



Step 3. Both paths must give the same dollars, or there is free money. So F = 1.10 × e^0.02 = 1.1222.



Step 4. Commodities: storage pushes forward prices up (contango); a convenience yield pulls them down (backwardation).

What it means

- **FX quote:** EUR/USD = 1.10 means 1 euro costs 1.10 dollars.
- **Covered interest rate parity:** you cannot earn free money by borrowing in one currency, converting, investing in the other, and locking in the rate back. So $F = S \cdot e^{(r - r_f)T}$.
- **FX exposure:** a US firm with euro revenue loses if the euro falls. Hedge by selling euros forward.
- **Commodities:** holding oil costs storage (u). But having it on hand also has a benefit, the **convenience yield** (y). $F = S \cdot e^{(r + u - y)T}$.
- **Contango**: F > S (storage dominates). **Backwardation**: F < S (high convenience yield, e.g. a shortage).
- **Equity risk** (also RMN3): value a stock with dividends or factors (CAPM, multi-factor).

The formulas

```
FX forward:        F = S × e^((r - r_f)·T)
Commodity forward: F = S × e^((r + u - y)·T)
```

S = spot price (in domestic currency per unit), r = domestic rate, r_f = foreign rate, u = storage cost rate, y = convenience yield.

Worked example

Question. EUR/USD spot is 1.10. USD rate 5%, EUR rate 3% (continuous). What is the 1-year forward? A US exporter will receive €1m in a year. What can it lock in?

1. F = 1.10 × e^((0.05 − 0.03) × 1) = 1.10 × e^0.02
2. e^0.02 = 1.0202 → F = `1.1222` USD per EUR
3. The euro is at a premium because its interest rate is lower.
4. Sell €1m forward: locks in `$1,122,200` no matter where the euro goes.

Answer: F = 1.1222; locks in $1.1222m

Calculator example

Inputs: Spot (USD per EUR) = 1.1, Domestic rate r (%) = 5, Foreign rate r_f (%) = 3, Years = 1, Foreign amount = 1000000

```
Forward = 1.1 × e^(0.0200 × 1) = 1.1222
Locked-in amount = 1,000,000 × 1.1222 = 1,122,221.47
```

Common traps

- Flipping r and r_f (check: the lower-rate currency is at a forward premium).
- Mixing up which currency the quote is "per".
- Leaving out convenience yield when the futures curve is downward sloping.

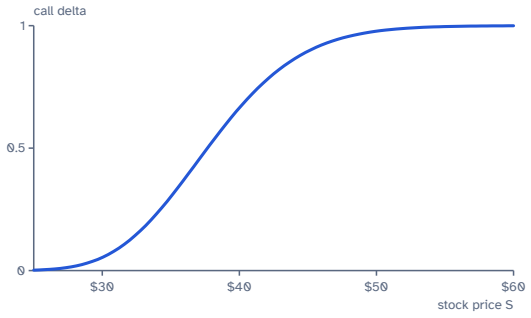
Where to read more

- Risk syllabus RMN3 (FX exposures and parity, equity, commodity carry and basis)
- Derivatives/03 Lectures/FRE6073_LECTURE_03_Lecture_02_Forward_and_Futures_Prices.pptx

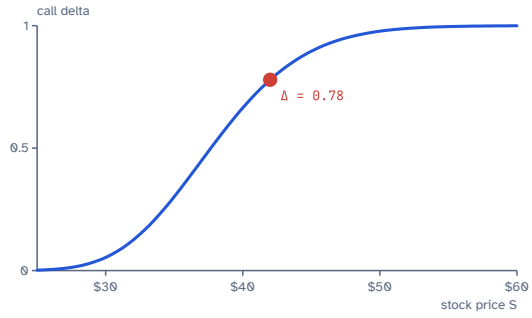
Greeks and delta hedging

A trader who sells options removes most of the risk by holding delta shares of the stock, then adjusting as delta changes. Gamma says how often you need to adjust.

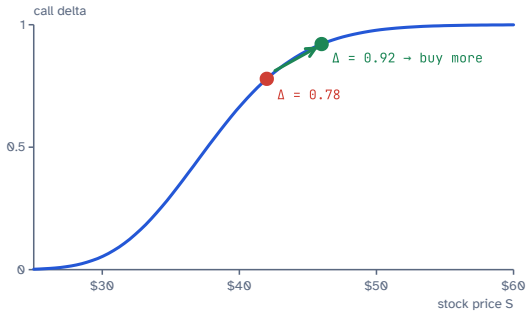
The sketch, step by step



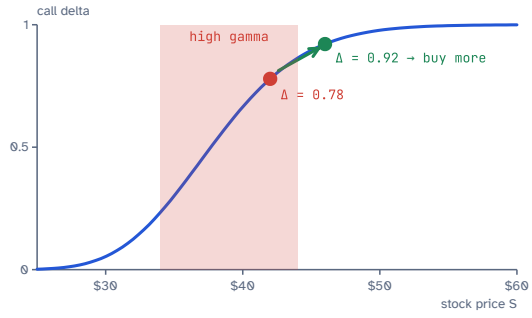
Step 1. Delta of a call (K = 40) against the stock price. It goes from 0 to 1.



Step 2. At S = 42 delta is about 0.78. Short 1 call → buy 0.78 shares.



Step 3. The stock rises to 46. Delta has climbed (that climb is gamma). You must buy more shares.



Step 4. The steepest part of the curve is near the strike: that is where gamma is highest and you rebalance most.

What it means

- **Delta-neutral:** total delta of the book = 0, so small stock moves do not change its value.
- Sold calls have negative delta, so you **buy** shares to hedge. Sold puts have positive delta, so you **sell** shares.
- **Gamma:** delta changes as the stock moves. New delta ≈ old delta + Γ × ΔS. High gamma = rebalance often.
- A short-option, delta-hedged book has negative gamma: it loses on big moves, and earns theta (time decay) on quiet days.
- **Gamma/vega hedging:** stock has zero gamma and vega, so you need other options to hedge those.
- **Hedging error:** you only rebalance at discrete times and volatility is not what you assumed, so the hedge is never perfect.

The formulas

```
Shares to hold = -(option position) × Δ
New Δ ≈ Δ + Γ × ΔS
P&L of hedged short option ≈ θ·Δt - ½·Γ·(ΔS)²   (per option, sign for a short)
```

Δ = delta, Γ = gamma, θ = theta (here per unit time, positive for a short option).

Worked example

Question. You sold 1,000 calls with delta 0.60 and gamma 0.04. How many shares to buy? The stock then rises $1.25. What do you do?

1. Option delta = −1,000 × 0.60 = −600 → buy `600 shares`
2. New delta ≈ 0.60 + 0.04 × 1.25 = `0.65`
3. You now need 1,000 × 0.65 = 650 shares
4. Buy `50 more`. (Notice: you buy after a rise and sell after a fall. That is how negative gamma costs money.)

Answer: Buy 600 shares, then 50 more

Calculator example

Inputs: Calls sold = 1000, Delta = 0.6, Gamma = 0.04, Stock move ($) = 1.25

```
Start: buy 600 shares
New delta ≈ 0.6 + 0.04 × 1.25 = 0.6500
Adjust: buy 50 shares
```

Common traps

- Selling shares to hedge a short call (it is buy).
- Thinking delta-neutral means no risk; gamma and vega remain.
- Forgetting the contract multiplier (1 contract often = 100 shares).

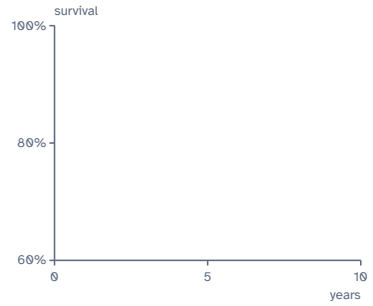
Where to read more

- Hull 8e, Chapter 18 (The Greek letters)
- Risk syllabus RMN4 (Greeks, dynamic hedging)

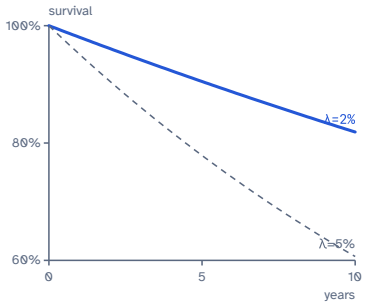
Credit risk: default, loss and spreads

Expected credit loss is chance of default × share you lose × amount at stake. A credit spread is the extra yield lenders demand to cover that.

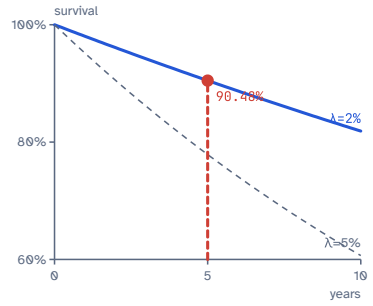
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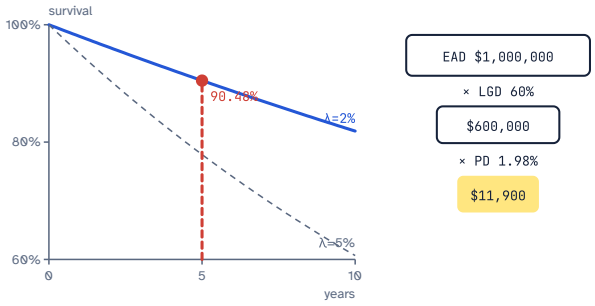
Step 1. The chance a borrower is still alive (not defaulted) over time.



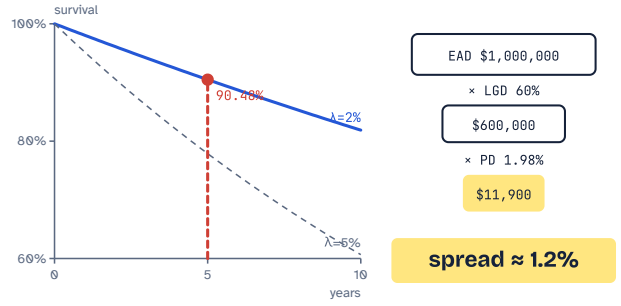
Step 2. With λ = 2% a year, survival slides slowly down: e^(−0.02t).



Step 3. At 5 years, 90.48% survive. So 9.52% have defaulted.



Step 4. Expected loss stacks three things: $1m at stake × 60% lost × 1.98% chance = $11.9k.



Step 5. The lender wants about λ × LGD = 1.2% extra yield to cover this.

What it means

- **PD** (probability of default), **LGD** (loss given default = 1 – recovery rate), **EAD** (exposure at default).
- **Hazard rate λ:** the default "speed". Survival to time t is $e^{-\lambda t}$; PD by t is $1 - e^{-\lambda t}$.
- **Credit spread ≈ λ × LGD.** A CDS (credit default swap) is insurance on a bond, and its spread works out to about the same number.
- **P vs Q:** real-world (P) default probabilities come from history. Risk-neutral (Q) ones come from prices and are higher, because they include a risk premium.
- **Expected loss** is a cost you price in. **Unexpected loss** is what capital must absorb.

The formulas

```
Survival:      S(t) = e^(-λt)        PD(t) = 1 - e^(-λt)
Expected loss: EL = PD × LGD × EAD
Spread:        s ≈ λ × LGD   →   λ ≈ s / (1 - R)
```

λ = hazard rate per year, R = recovery rate.

Worked example

Question. A borrower has λ = 2% a year. LGD = 60%. Exposure $1m. Find 5-year survival, 5-year PD, 1-year expected loss and the spread.

1. Survival 5y = e^(−0.02 × 5) = e^(−0.1) = `90.48%`
2. PD 5y = 1 − 0.9048 = `9.52%`
3. 1-year PD ≈ 1 − e^(−0.02) = 1.98%, EL ≈ 0.0198 × 0.6 × 1,000,000 = `$11,900`
4. Spread ≈ 0.02 × 0.6 = `1.2%` = 120 basis points

Answer: Survival 90.48%, PD 9.52%, EL ≈ $11.9k, spread ≈ 120 bp

Common traps

- Using recovery rate where LGD is needed (or the reverse).
- Adding PDs across years instead of using survival.
- Mixing P and Q probabilities in the same calculation.

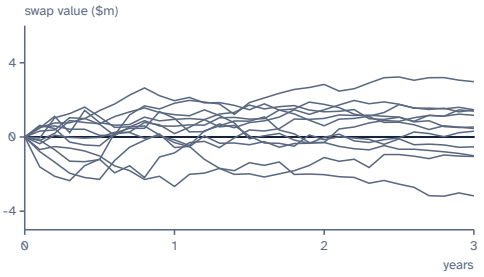
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_10_Lecture_03_Corporate_Bonds_and_CDOs.pdf`
- `Risk syllabus RMN5-6 (PD, LGD, CDS, CVA)`
- `Hull 8e, Chapter 23 (Credit risk)`

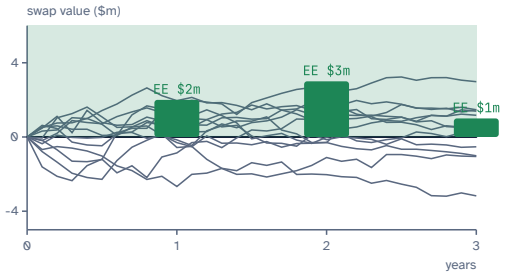
Counterparty risk and CVA

If the other side of a derivative defaults while it owes you money, you lose. CVA is the price of that risk, built from exposure, default probability and loss given default.

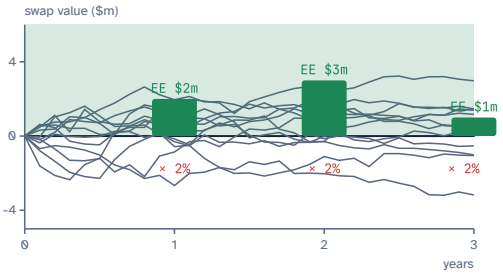
The sketch, step by step



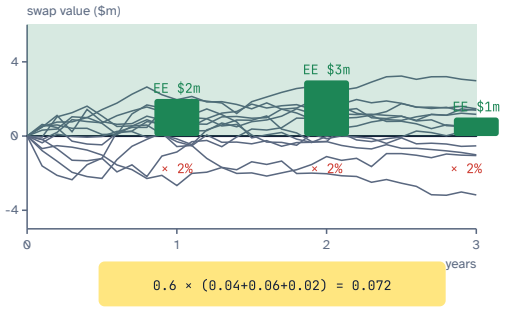
Step 1. Possible future values of a swap. Some paths end up in your favour (above 0), some against you.



Step 2. Only the part above zero is at risk. Averaging that gives the expected exposure: a hump over time.



Step 3. Multiply each year's exposure by the chance of default in that year.



Step 4. Add them up and multiply by LGD: CVA ≈ $72k.

What it means

- **Exposure:** only positive value matters. If the trade is worth +5 to you and they default, you lose up to 5. If it is worth −5, you still owe them.
- **Expected exposure EE(t):** the average of max(V, 0) across simulated futures. For a swap it rises then falls (a hump), because fewer payments remain later.
- **PFE (potential future exposure):** a high percentile (e.g. 95%) of exposure. Used for limits.
- **CVA (credit valuation adjustment):** expected loss from counterparty default, subtracted from the trade value.
- **Reducing it:** netting, collateral (margin), and central clearing. xVA adds related adjustments (DVA, FVA, KVA).
- **Migration:** a counterparty can be downgraded before defaulting, which raises CVA too.

The formulas

```
Exposure:  E(t) = max(V(t), 0)
CVA ≈ LGD × Σ EE(t_i) × PD(t_{i-1}, t_i) × DF(t_i)
```

PD(t_{i−1}, t_i) = probability of default in that period, DF = discount factor.

Worked example

Question. A 3-year swap has expected exposure $2m, $3m, $1m in years 1, 2, 3. The counterparty has a 2% chance of defaulting in each year; LGD 60%. Ignore discounting. CVA?

1. Year 1: 2 × 0.02 = `0.04`
2. Year 2: 3 × 0.02 = `0.06`; Year 3: 1 × 0.02 = `0.02`
3. Sum = 0.12, times LGD 0.6 = `0.072`
4. CVA ≈ $72,000. The trade is worth $72k less than if the counterparty were risk-free.

Answer: CVA ≈ $72,000

Common traps

- Using V instead of max(V, 0).
- Using cumulative PD for every year instead of the PD within each period.
- Forgetting that collateral and netting cut exposure.

Where to read more

- Risk syllabus RMN5-6 (CVA, exposure profiles, xVA)
- Hull 8e, Chapter 24 (Credit risk)

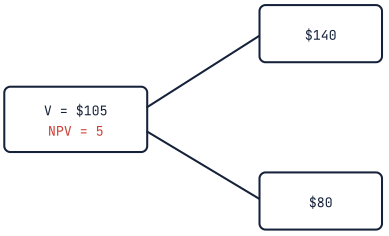
Real options and the Merton model

Choices in a business (wait, expand, abandon) are options, so they have value that plain NPV misses. Even a company's stock is an option on its assets.

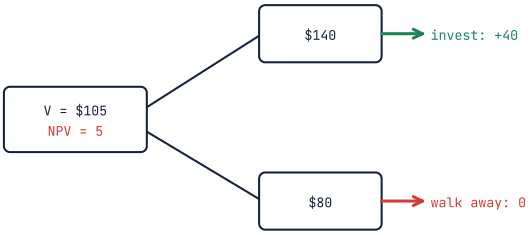
The sketch, step by step



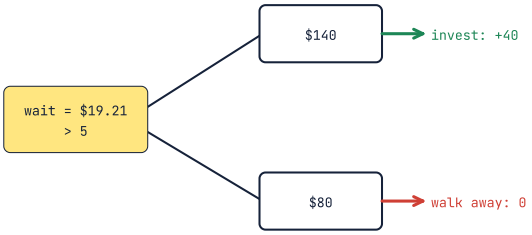
Step 1. Today the project is worth $105 and costs $100. NPV = $5.



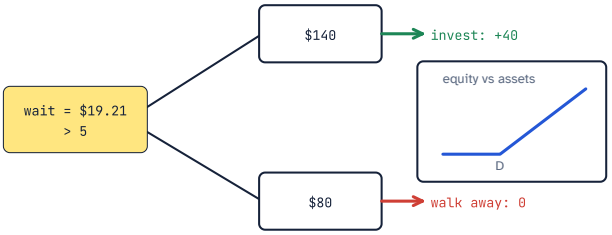
Step 2. Next year it is worth $140 (good) or $80 (bad).



Step 3. If you wait, you invest only in the good case: gain $40. In the bad case you walk away: $0.



Step 4. Rolled back with p = 0.504: the right to wait is worth $19.21. Waiting wins.



Step 5. Same idea for a firm: shareholders get max(Assets – Debt, 0). Equity is a call on the assets.

What it means

- **Real option:** the right, not the duty, to take a business action later. Waiting lets you invest only if things go well.
- Value it like Class 16: build a tree of the project value, find risk-neutral p, roll back.
- If the option value is bigger than investing now (NPV today), **wait**.
- **Energy example (VL14):** a power plant is a call on the spread between power and fuel prices: run it only when profitable.
- **Merton model:** equity = call option on the firm's assets with strike = debt. Default happens if assets end below the debt. **Distance to default** counts how many σ the firm is from that line.

The formulas

```
Risk-neutral p = [V0·(1 + rf) - V_down] / (V_up - V_down)
Option value   = [p·max(V_up - I, 0) + (1-p)·max(V_down - I, 0)] / (1 + rf)
Merton:  Equity = V·N(d1) - D·e^(-rT)·N(d2)
         DD = [ln(V/D) + (μ - σ²/2)T] / (σ√T),   PD = N(-DD)
```

V = value of project or firm assets, I = investment cost, D = face value of debt.

Worked example

Question. A project is worth $105 today and costs $100. Next year it will be worth $140 or $80; rf = 5%. Invest now, or keep the right to invest next year (cost still $100)?

1. NPV now = 105 − 100 = `5`
2. p = (105 × 1.05 − 80) / (140 − 80) = 30.25/60 = `0.504`
3. Wait: next year invest only if up: payoff 40 or 0
4. Option value = 0.504 × 40 / 1.05 = `19.21`. Bigger than 5, so wait.
5. Merton check: V = 100, D = 80, μ = 6%, σ = 25%, T = 1 → DD = (0.2231 + 0.0288)/0.25 = `1.01`, PD = N(−1.01) ≈ `15.7%`

Answer: Waiting is worth $19.21 vs $5 now. Merton PD ≈ 15.7%

Common traps

- Using the real-world probability in the roll-back.
- Comparing the option with zero instead of with investing now.
- In Merton, using the stock's volatility as the asset volatility.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_14_VL14_Real_Options_in_Energy.pdf`
- `Valuation/02 Books and Readings/FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf`
- `Risk syllabus RMN6 (Merton, distance to default)`

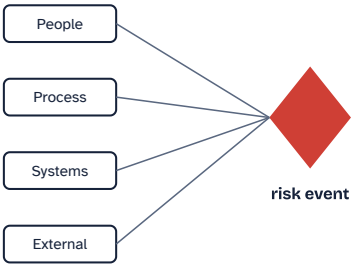
Operational risk and controls

Operational risk is loss from failed people, processes, systems or outside events: a fat-finger trade, a hack, a payment sent twice, fraud.

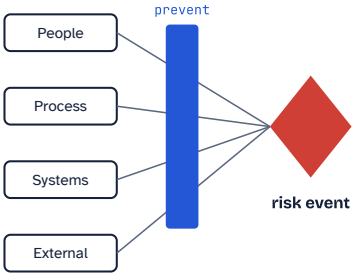
The sketch, step by step



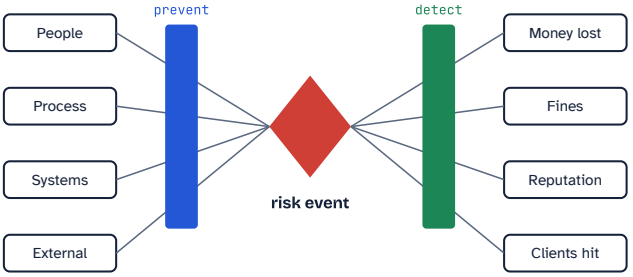
Step 1. A bow-tie: the risk event sits in the middle (e.g. a payment sent twice).



Step 2. On the left are the causes: people, process, systems, external events.



Step 3. Preventive controls sit between causes and the event. They try to stop it.



Step 4. On the right are the consequences. Detective controls and recovery plans limit the damage.

What it means

- **Basel definition:** loss from inadequate or failed internal processes, people and systems, or from external events. Includes legal risk.
- **Frequency × severity:** many small losses (errors) and rare huge ones (fraud, rogue trading). The rare ones dominate the tail.
- **Controls:** preventive (stop it happening: approvals, access limits) and detective (catch it: reconciliations, alerts).
- **KRIs (key risk indicators):** early-warning numbers, e.g. failed trades per day, staff turnover, overdue reconciliations.
- **Three lines of defense:** 1) the business owns the risk, 2) risk & compliance oversee, 3) internal audit checks independently.
- **Payments and settlement risk:** paying before you get paid (Herstatt risk). Fixed by delivery-versus-payment and CLS for FX.
- **Technology/cyber risk** and model risk sit here too.

The formulas

```
Expected annual loss ≈ expected frequency × expected severity
Residual risk = inherent risk - effect of controls
```

Op-risk capital looks at the tail of the loss distribution, not just the average.

Worked example

Question. A desk has about 12 settlement errors a year, each costing $50,000 on average. A new reconciliation control would catch 75% of them and costs $200,000 a year. Is it worth it (just on expected loss)?

1. Expected loss now: 12 × 50,000 = `$600,000` a year
2. Losses avoided: 75% × 600,000 = `$450,000`
3. Net benefit: 450,000 − 200,000 = `$250,000` a year
4. Yes, and it also cuts tail risk (fewer chances of one giant error).

Answer: Worth it: saves about $250,000 a year

Common traps

- Looking only at average losses and ignoring rare severe events.
- Confusing a control (an action) with a KRI (a measurement).
- Treating operational risk as only IT risk.

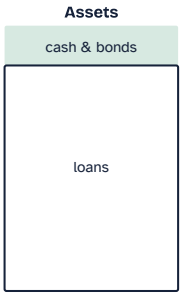
Where to read more

- Risk syllabus RMN7 (operational risk, payments, controls, KRIs, technology risk)

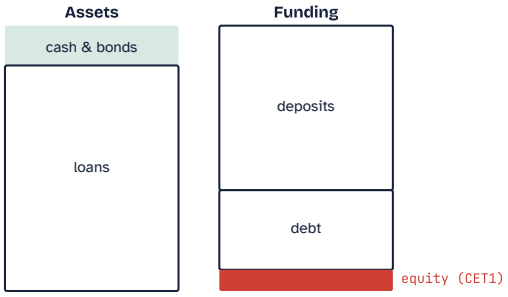
Bank capital and liquidity rules

Banks must keep enough capital (their own money) to absorb losses and enough liquid assets to survive a month of panic withdrawals. Basel III sets the minimums.

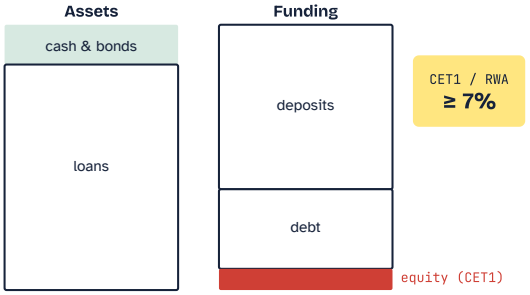
The sketch, step by step



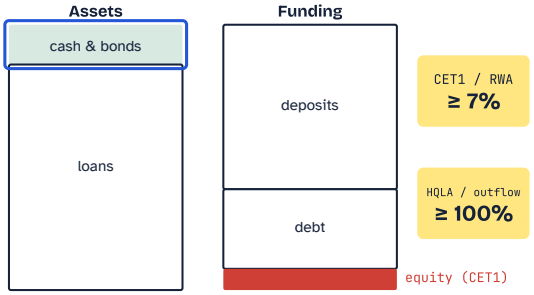
Step 1. A bank balance sheet: assets on the left, how they are funded on the right.



Step 2. Most funding is deposits and debt. A thin slice is equity: that is capital.



Step 3. Weight the assets by risk to get RWA. Capital must be at least 7% of that (with the buffer).



Step 4. Liquidity is a separate test: enough liquid assets to cover 30 days of stressed outflows.

What it means

- **Capital:** equity that can absorb losses. CET1 (common equity tier 1) is the best quality.
- **Risk-weighted assets (RWA):** each asset is weighted by riskiness (cash ~0%, mortgages lower, corporate loans ~100%). Includes market and operational risk too.
- **Ratios (Basel III minimums):** CET1 ≥ 4.5% of RWA, Tier 1 ≥ 6%, total capital ≥ 8%, plus a 2.5% conservation buffer (and extra for big banks).
- **Leverage ratio:** Tier 1 / total exposure (not risk weighted) ≥ 3%. A backstop if risk weights are wrong.
- **LCR (liquidity coverage ratio):** high-quality liquid assets / net cash outflows over 30 days of stress ≥ 100%.
- **NSFR (net stable funding ratio):** stable funding / required stable funding ≥ 100% over one year.

The formulas

```
CET1 ratio = CET1 / RWA              (≥ 4.5%, ≥ 7% with buffer)
Leverage   = Tier 1 / Total exposure  (≥ 3%)
LCR        = HQLA / 30-day net outflows (≥ 100%)
```

HQLA = high-quality liquid assets (cash, central bank reserves, government bonds).

Worked example

Question. A bank has CET1 $60bn, RWA $800bn, total exposure $1,500bn, HQLA $120bn and stressed 30-day outflows of $100bn. Assume Tier 1 = CET1. Does it pass?

1. CET1 ratio = 60 / 800 = `7.5%` ≥ 7% ✓
2. Leverage = 60 / 1,500 = `4.0%` ≥ 3% ✓
3. LCR = 120 / 100 = `120%` ≥ 100% ✓
4. It passes all three, but only 0.5% above the CET1 buffer: a $4bn loss would breach it.

Answer: 7.5% CET1, 4% leverage, 120% LCR: passes

Calculator example

Inputs: CET1 ($bn) = 60, RWA ($bn) = 800, Total exposure ($bn) = 1500, HQLA ($bn) = 120, 30-day outflows ($bn) = 100

```
CET1 ratio = 7.50% ✓ (need 7%)
Leverage = 4.00% ✓ (need 3%)
LCR = 120% ✓ (need 100%)
```

Common traps

- Dividing capital by total assets for the CET1 ratio (use RWA).
- Thinking capital is cash. Capital is how the bank is funded, not what it holds.
- Mixing up liquidity (cash to pay today) and solvency (enough capital overall).

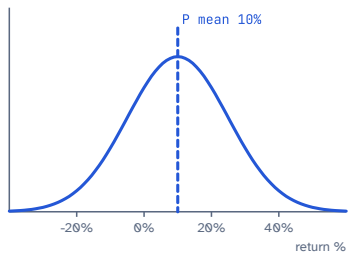
Where to read more

- Risk syllabus RMN9 (bank balance sheet, capital, leverage, liquidity)
- BIS Basel Framework (see "Watch online")

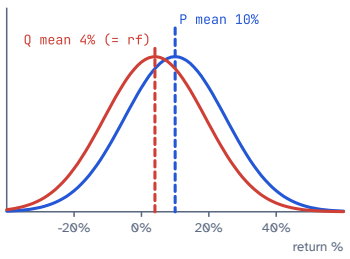
P vs Q, risk drivers and stress tests

P is the real world (what actually tends to happen). Q is the risk-neutral world used for pricing. Risk managers measure losses in P, price in Q, and stress-test both.

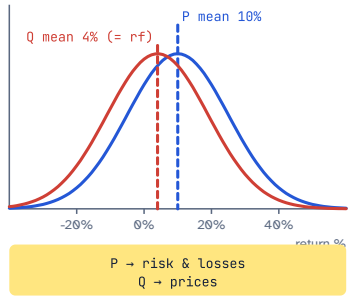
The sketch, step by step



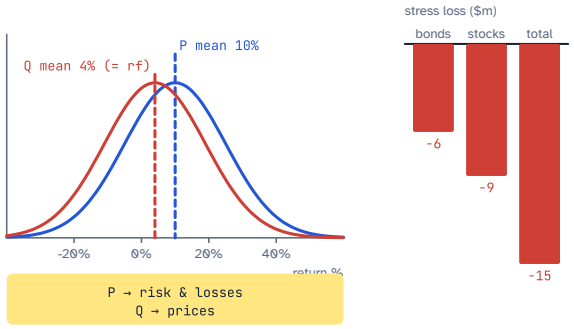
Step 1. P: the real-world spread of outcomes for a risky asset. Its average is above the risk-free rate.



Step 2. Q: the same outcomes reweighted so the average is the risk-free rate. More weight on bad outcomes.



Step 3. Use P for risk (VaR, expected loss). Use Q for prices.



Step 4. Stress test: shock each driver and add the hits. Bonds –$6m, equities –$9m, total –$15m.

What it means

- **Q (risk-neutral):** probabilities that make every asset earn the risk-free rate. Used for pricing (binomial trees, Black-Scholes, CDS). Not a forecast.
- **P (physical/real world):** probabilities from history and judgement. Used for VaR, expected losses, capital.
- Q puts more weight on bad outcomes because investors dislike them. That is why Q default probabilities from spreads are higher than historical ones.
- **Risk drivers:** the few market factors that move a portfolio (rates, spreads, equity index, FX, vol). Map each position to them.
- **Aggregation:** combine risks across desks using correlations; diversification lowers the total, but correlations jump in a crisis.
- **Ex-ante P&L / stress tests:** apply a scenario (e.g. rates +200bp, equities –30%, spreads +150bp) and reprice everything to see the loss before it happens.

The formulas

```
Q default intensity from spread:  λQ ≈ s / LGD
Stress loss ≈ Σ (sensitivity to driver × shock)    (or full revaluation)
Bond:  ΔV ≈ -V × D × Δy
```

Use P for "how likely / how much could we lose", Q for "what is it worth".

Worked example

Question. A bond portfolio worth $50m (duration 6) and an equity portfolio worth $30m. Stress: rates +2%, equities −30%. Loss? Also: a spread of 150bp with 60% LGD implies what Q default rate, if history says 1%?

1. Bonds: −50 × 6 × 0.02 = `-$6m`
2. Equities: −30% × 30 = `-$9m`
3. Total stress loss ≈ `$15m` (19% of $80m)
4. λQ ≈ 0.015 / 0.6 = `2.5%` a year vs 1% in P. The gap is the credit risk premium.

Answer: Stress loss ≈ $15m; Q default rate 2.5% vs P 1%

Common traps

- Using Q probabilities to forecast real default rates.
- Adding stress losses that cannot happen together, or ignoring ones that do.
- Trusting normal-times correlations in a stress scenario.

Where to read more

- `Risk syllabus RMNA (P vs Q, drivers, aggregation, ex-ante P&L, stress)`
- `Valuation notes: instructor-linked explanation of risk neutrality`

Watch and learn online

Free videos and courses to use alongside these classes. Coursera courses are free when you choose "audit".

Risk Management

- **Bionic Turtle (YouTube)** (Free FRM-style videos)
  https://www.youtube.com/@bionicturtle
  Best free match for this course: VaR vs expected shortfall, duration, credit risk, CVA, operational risk and Basel. Search the channel by topic.
- **TU Delft OpenCourseWare: Special VaRs and Expected Shortfall** (Free university video lecture)
  https://ocw.tudelft.nl/course-lectures/3-2-special-vars-expected-shortfall/
  A clear short lecture on VaR and ES with pictures.
- **Columbia (Coursera): Introduction to Financial Engineering and Risk Management** (Free to audit, video course)
  https://www.coursera.org/learn/financial-engineering-intro
  Term structure, credit and risk-neutral pricing (P vs Q).
- **MIT OpenCourseWare 18.S096: Mathematics with Applications in Finance** (Free full lecture videos)
  https://ocw.mit.edu/courses/18-s096-topics-in-mathematics-with-applications-in-finance-fall-2013/
  Lectures on portfolio theory, time series, credit and risk measures, taught by practitioners.
- **BIS: The Basel Framework** (Official rules (reading, not video))
  https://www.bis.org/basel_framework/
  The source for capital, leverage and liquidity ratios. Use it to check numbers, not to learn from scratch.